

CE 251: Hydraulics Engineering

Course Code	CE 251
Course Name	Hydraulics Engineering
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	UG
Distribution	Discipline Core
Approval	9th Senate (May 2016)

Course Contents

- **Introduction**
definition of fluid, liquids and gases, continuum hypothesis, Newtonian and non-Newtonian fluids [6 Lectures]
- **Fluid Statics**
Pascal's law, Manometry, Buoyancy, metacentric height, rigid body motion. [6 Lectures]
- **Fluid Kinematics**
Lagrangian and Eulerian fluid motion, vorticity and circulation, rotational and irrotational flows. [6 Lectures]
- **Fluid Dynamics:**
Reynolds transport theorem, Equation of mass momentum and inertia, Integral formulation of governing equations, Euler's equation, Bernoulli's equation, Navier-Stokes equation. [8 Lectures]
- **Internal Flows**
Couette Flow, Hagen-Poiseuille flow, flow through pipe, channels, Venturi, Orifice, head loss calculations, Moody's chart. [8 Lectures]
- **Dimensional Analysis**
scaling and similarity, Buckingham π – theorem, model testing. [4 Lectures]
- **External Flows**
Boundary layer flows, laminar and turbulent flows, flow separation, lift and drag, Stokes Law, displacement and momentum thickness. [4 Lectures]
- **Flow in open channel**
Discharge measurements in open channel, Concept of Specific Energy, Critical flow and depth computation, application of specific energy. [2 Lectures]
- **Introduction to hydrology**
Meteorological cycle, rainfall and runoff calculation. [1 Lecture]

Text Books

1. Hunter Rouse, **Elementary Mechanics of Fluids**, John Wiley & Sons, 2011.
2. V.L. Streeter and E.B. Wylie, **Fluid Mechanics**, McGraw Hill Book Co., 1962.
3. P.N. Modi and S.M. Seth, **Hydraulics & Mechanics**, Standard Book House, 2002.
4. B. S. Massey, **Mechanics of Fluids**, Van Nostrand Reinhold Co., 1979.
5. J. Frabzini, **Fluid Mechanics with Engineering Applications**, McGraw Hill, 1997.
6. J.H. Spurk, **Fluid Mechanics – Problems and Solutions**, Springer, 2003.
7. Wilson, E. M., **Engineering hydrology (Vol. 4)**, Macmillan, 1990.
8. Subramanya, K., **Engineering hydrology**, Tata McGraw-Hill Education, 1994.

References

1. Lewitt, Ernest H., **Hydraulics and Fluid Mechanics: A Text-book Covering the Syllabuses of the B. Sc.(Eng.), ICE and I. Mech. E. Examinations in this Subject**, Pitman, 1958.
2. Dixon, S. L., and Hall, C., **Fluid mechanics and thermodynamics of turbomachinery**, Butterworth-Heinemann, 2013.